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AGBUF User's Manual

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AGBUF User's Manual:

Introduction:

Program “AGBUF” (**A**GNPS **B**uffer **U**tility **F**eature) is a module in the suite of modules called TopAGNPS. It was written to allow the user to create an input buffer data section for use with AnnAGNPS. The goal of this program is to produce a riparian buffer input file for AnnAGNPS that contains information about the buffer such as its location within the AGNPS cells or reaches, vegetative type, slope, width, maximum trapping efficiency, and drainage area to the buffer. It can also calculate the maximum trapping efficiency for ephemeral gullies if the output file from the Potential Ephemeral Gully (PEG) is included as input to AGBUF.

Most of the input files for AGBUF, as described below, can be generated by executing the following modules: DEDNM, RASPRO, RASFOR, AgFlow, and optionally PEG. The TopAGNPS modules, (DEDNM, RASPRO, and RASFOR), produces spatial files required by AGBUF. AgFlow produces two spatial output files called “AnnAGNPS_Cell_IDs.asc” and “AnnAGNPS_Reach_IDs.asc” which are also required by AGBUF. PEG produces an output file that can be modified by AGBUF to provide a maximum trapping efficiency for ephemeral gullies.

There are two required spatial input files that must be supplied but are not produced by the programs listed above: 1.) a buffer zone file; and 2.) a vegetation type file. These files are described below.

AGBUF produces several output files which are described below. The main output file of interest is the “AnnAGNPS_Riparian_Buffer_Data_Section_AgBuf.csv” output file. This is the file that may be used directly by AnnAGNPS as the “Riparian Buffer Data” section.

Input files:

There are eight required and two optional input files, described below, necessary for the successful execution of AGBUF. Of the eight required input files, the first is a control file and the remaining seven are spatial layer files. Five of the seven layer files are specifically named and the remaining two are user-defined. Also, if executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

Control file:

AGBUF.CSV or AGBUF.XML

The required control file allows the user to specify a number of parameters that affect the processing of spatial data and the determination of riparian buffer locations. If both “AGBUF.CSV” and

“AGBUF.XML” are found, the “AGBUF.CSV” control file will be processed. A message will be produced in the log file and AGBUF execution terminated if the control file is not present.

There are two possible formats that may be used; csv-formatted or xml-formatted.

“AGBUF.CSV” is a csv-formatted control file. The first record in the control file is the header record containing at least one of several possible keywords that may be used. The second record is the data record containing the values for the keywords that are specified in the header record. Only two records are read, header and data; therefore, any other information contained within the control file is ignored. The csv-formatted control file must be named “AGBUF.CSV”. An example is listed below in the “Examples” section.

“AGBUF.XML” is xml-formatted control file. If the xml-format is used, it must adhere strictly to the following –only one tag per line, with no spaces. The xml-formatted control file must be named “AGBUF.XML”. An example is listed below in the “Examples” section.

Keywords:

All parameters may be present but only required parameters and any desired optional parameters need be present. The keyword lines above are listed in alphabetical order but may be entered in any order. All keywords must be spelled exactly as specified below. The keywords are case and order insensitive.

“BUFFER”

This optional keyword record allows the user to specify the name and location of the spatial buffer layer that is to be used. For example, “my_buffer.asc”. If the filename is not included then AGBUF will attempt to use a default of “buffer.asc”. If the buffer file is not found then an error message will be produced.

“VEGETATION”

This optional keyword record allows the user to specify the name and location of the spatial vegetation layer that is to be used. For example, “my_veg.asc”. If the filename is not included then AGBUF will attempt to use a default of “vegetation.asc”. If the vegetation file is not found then an error message will be produced.

“FOREST”

This optional keyword record allows the user to designate the value that was assigned as forest in the spatial vegetation layer. For example, “11”, indicates that any raster with the value of “11” is to be treated as forest. Default = -99

“GRASS”

This optional keyword record allows the user to designate the value that was assigned as grass in the spatial vegetation layer. For example, “10”, indicates that any raster with the value of “10” is to be treated as grass. Default = -99

“C_THRESHOLD”

This optional keyword record allows the user to specify a threshold value for the cells in number of rasters. If the drainage area number of rasters along a particular flow-path into the buffer is larger than this threshold value then the trapping efficiency is set to 0.0 for that flow-path. For example, “40”, indicates that any flow-path entering the buffer that is larger than “40” rasters will have no trapping efficiency, “0.0”. Default is that no threshold is applied.

“R_THRESHOLD”

This optional keyword record allows the user to specify a threshold value for the reaches. This allows the user to specify a minimum value for buffer width that is required to be included in a reach-located buffer. For example, “10 . 0”, indicates that the buffer width at a given point in the reach must be at least 10.0 meters wide on both sides of the reach to be included in the reach-located buffer. The rasters that are checked are those that extend perpendicular to the flow-vector of the reach raster in both directions. The default reach threshold value is $\frac{1}{2}$ of the grid's raster width.

“ASC_PATH”

This optional keyword allows the user to specify the path to where the spatial files created by DEDNM, RASPRO, RASFOR, and AgFlow are located. This path will also be used for spatial output files created by AGBUF.

This keyword is optional and if not included then the default location of the spatial input files and AGBUF spatial output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the required spatial input files and AGBUF spatial output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the spatial input files and AGBUF spatial output files.

NOTE: This keyword is only used for independent runs of the AGBUF module. It is ignored when AGBUF is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from AGBUF will be placed in the default folder location.

“CSV_PATH”

This optional keyword allows the user to specify the path to where any csv-formatted input files and the csv-formatted output files created by AGBUF will be written.

This keyword is optional and if not included then the default location of any csv-formatted input files and AGBUF csv-formatted output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for any csv-formatted input files and AGBUF csv-formatted output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for any csv-formatted input files and AGBUF csv-formatted output files.

NOTE: This keyword is only used for independent runs of the AGBUF module. It is ignored when AGBUF is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from AGBUF will be placed in the default folder location.

“UNITS”

This optional keyword record allows the user to specify the output units, SI or English, for the “AGBUF_AnnAGNPS.csv” file. A value of “1” indicates SI units and a value of “0” indicates English units. For example, “1” indicates that the output will be in SI units (m, m2). If “0” then output will be in English units (ft, ft2). Default value is 1 (SI).

Example

The following is an Excel example of the csv-formatted control file showing all available keywords:

Buffer	Vegetation	Forest	Grass	C_Threshold	R_Threshold	ASC_PATH	CSV_PATH	Units
Buffer .asc	Vegetation .asc	-99	2	4000.	10.	./Output_Files/ ASC	./Output_Files/ CSV	1

The following is the same information displayed within a text editor showing all available keywords:

```
Buffer,Vegetation,Forest,Grass,C_Threshold,R_Threshold.,ASC_PATH,CSV_PATH,Units
buffer.asc,vegetation.asc,-99,2,4000.,10.,./Output_Files/ASC,./Output_Files/CSV,1
```

The following is an example of the xml-formatted control file:

```
<TOPAGNPS>
<KEYWORD ID="ASC_PATH">./Output_Files/ASC</KEYWORD>
<KEYWORD ID="BUFFER">Buffer.asc</KEYWORD>
<KEYWORD ID="C_THRESHOLD">4000.</KEYWORD>
<KEYWORD ID="CSV_PATH">./Output_Files/CSV</KEYWORD>
```

```
<KEYWORD ID="FOREST">-99</KEYWORD>  
<KEYWORD ID="GRASS">2</KEYWORD>  
<KEYWORD ID="R_THRESHOLD">3.</KEYWORD>  
<KEYWORD ID="UNITS">1</KEYWORD>  
<KEYWORD ID="VEGETATION">Vegetation.asc</KEYWORD>  
</TOPAGNPS>
```

Input data files:

There are eight input data files required for AGBUF and two optional input files. Seven of the required input data files are spatial layers as described below. The spatial layers must have the same dimensions with the same header information in each file. All of the filenames must be as specified below with the exception of the buffer layer and vegetation layer. Also, if executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

Required input files:

AnnAGNPS_Cell_IDs.asc (AgFlow):

This required spatial input file is the subwatershed (AGNPS cell) designations layer.

AnnAGNPS_Reach_IDs.asc (AgFlow):

This required spatial input file is the reach (AGNPS reach) designations layer.

Buffer.asc (or other user-specified filename):

This required spatial layer input file specifies where the buffer zones are located with a required value of "2". This file must be of the same dimension with the same header information as the other spatial input files produced by TopAGNPS.

FLOVEC.ASC (TopAGNPS):

This required spatial input file is the flow vector layer.

TSLOPE.ASC (TopAGNPS):

This required spatial input file is the slope layer.

UPAREA.ASC (TopAGNPS):

This required spatial input file is the drainage area layer.

Vegetation.asc (or other user-specified filename):

This required spatial input file specifies the vegetation layer. There are currently only 2 types of vegetation recognized and processed by AGBUF; forest and grass. The vegetation layer can have separate user-defined values for each. This file must be of the same dimension with the same header information as the other spatial input files produced by TopAGNPS.

Optional input files:

PEG_Summary.csv (PEG):

AnnAGNPS_Ephemeral_Gully_Data_Section.csv (PEG):

There are two optional csv-formatted input data files that may be used with AGBUF. The two files are the output files from the Potential Ephemeral Gully (PEG) program with the following names respectively: “PEG_Summary.csv” and “AnnAGNPS_Ephemeral_Gully_Data_Section.csv”.

If “PEG_Summary.csv” and “AnnAGNPS_Ephemeral_Gully_Data_Section.csv” exist in the folder where the required input files for AGBUF reside, then they are assumed to be used as input by AGBUF. If used, both must be present.

AGBUF will read the list of cell-located and reach-located peg points from the “PEG_Summary.csv” input file and begin processing. AGBUF first identifies peg points that fall within AGNPS cells and reaches that also contain one or more buffers as identified by AGBUF.

If a peg point is located within a cell or reach that also contains a buffer, then the location of the peg point is checked to see if the peg point is inside or outside of the buffer zone.

If outside the buffer, then the flow path, starting with the peg point location, is traversed to see if it enters the buffer. If the flow path enters and exits the buffer, the trapping efficiency of the flow path is written to the appropriate parameter field in the “AnnAGNPS_Ephemeral_Gully_Data_Section.csv” file. Also the drainage area to the buffer replaces the current value for the “Cell’s drainage subcell” parameter for cell-located gullies.

If inside the buffer, then the term “BUFFER” is written to the management field ID for the gully in the “AnnAGNPS_Ephemeral_Gully_Data_Section.csv” file. AnnAGNPS will view this as indicating that the gully is ineffective and no gully sediment erosion or yield will be produced; effectively disabling the gully.

Output files:

There are currently nine output files generated by the program; there are ten if the PEG input files are used as described above.

Log file:

The log file, "AGBUF.log", is automatically generated and is a log of the processing as the program continues its execution. Additionally, the log files for "TopAGNPS" will also contain information specific for AGBUF. The log file, as it is being written, is also displayed to the screen. The log file contains processing information as well as error and warning messages.

Intermediate Output data files:

AgBuf_Buffer_Zone_Cell_Edge.asc:

This is a computed intermediate spatial layer that assigns a value of "1", "2", or "3" to buffer rasters in the cells that meet certain criteria. A "1" indicates that it is an upstream-edge raster meaning that it is the first raster in the buffer along a flow-path that receives flow from outside of the buffer. A value of "2" indicates that it is a downstream-edge raster meaning that it is the last buffer raster along a flow-path before entering a reach. A value of "3" indicates that the raster is designated as both, an upstream-edge and downstream-edge raster.

AgBuf_Buffer_Zone_Edge_Rasters.asc:

This is a computed intermediate spatial layer that assigns a value of "1", "2", or "3" to buffer rasters in the cells that meet certain criteria. A "1" indicates that it is an upstream-edge raster meaning that it is the first raster in the buffer along a flow-path that receives flow from outside of the buffer. A value of "2" indicates that it is a downstream-edge raster meaning that it is the last buffer raster along a flow-path before entering a reach that received flow from one or more upstream-edge rasters. A value of "3" indicates that the raster is a downstream-edge raster with flow that originated within the buffer; that is, there is no associated upstream-edge raster.

AgBuf_Buffer_Zone_Reach_Edge.asc:

This is a computed intermediate spatial layer that assigns a value of "1", "2", or "3" to buffer rasters in the reaches that meet certain criteria. A "1" indicates that it is an upstream-edge raster meaning that it is the first raster in the buffer in the reach flow-path that receives flow from outside of the buffer. A value of "2" indicates that it is a downstream-edge raster meaning that it is the last buffer raster in a reach flow-path before either exiting the buffer zone or the reach. A value of "3" indicates that the raster is designated as both, an upstream-edge and downstream-edge raster.

AgBuf_Catchment_Area.asc:

This is a computed intermediate spatial layer that contains the drainage area values at any given raster location.

AgBuf_Reach_Located_Buffers.asc:

This is a computed intermediate spatial layer that assigns a value of “1” or “2”. First, a value of “1” is assigned to all reach rasters that are located in a buffer zone. Then reach rasters that meet the criteria are assigned a value of “2”. The criteria are, 1.) The buffer must extend on both sides of the reach, and 2.) The buffer-width must meet the buffer-width threshold. This file is used to compute the “buffer_zone_reach.asc” file described above.

Output data files:**AgBuf_Cell-Reach_Buffer_Flow_Path_Summary.csv:**

This output file is a csv file that contains expanded information about each flow-path in the cells and reaches.

AnnAGNPS_Riparian_Buffer_Data_Section_AgBuf.csv:

This output file is a csv file that contains information that can be imported as the “Riparian Buffer Data” section into the AGNPS Input Editor or used directly by AnnAGNPS if used with the AnnAGNPS csv input option.

AnnAGNPS_Ephemeral_Gully_Data_Section_Revised_by_AgBuf.csv:

This output file is a csv file that potentially contains modified information for the “AnnAGNPS_Ephemeral_Gully_Data_Section.csv” file that was produced by the PEG program. This file can be imported as the “Ephemeral Gully Data” section into the AGNPS Input Editor or used directly by AnnAGNPS if used with the AnnAGNPS csv input option.